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IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-225

1. Topic (12 words max)

AI-Driven Health Analyzer Real-Time Physical and Mental Wellness Optimization using ML

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-225
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The increasing prevalence of lifestyle diseases, mental health disorders, and environmental stressors has created a demand for personalized, data-driven health solutions. While health monitoring technologies have advanced, many current systems still rely heavily on specialized sensors or fail to offer real-time, holistic optimization for both physical and mental health. Most existing health solutions focus on isolated aspects, such as hydration or fitness, without addressing the interconnectedness of mental health, environmental factors, and personalized recommendations.

A key limitation is the dependence on wearable devices, which may not be accessible or desirable for all users. Furthermore, many systems lack a comprehensive, adaptive approach to health, forcing individuals to use multiple apps or gadgets for different health domains. There is a pressing need for an AI-powered health management system that adjusts to users' changing needs while minimizing reliance on physical sensors.

This research aims to develop a comprehensive AI-driven health tool that assesses hydration, provides feedback, and detects the correct position for exercising. Time detects emotional cues and integrates environmental data to offer personalized health recommendations. The system uses machine learning models to provide real-time, dynamic health optimization. This approach is particularly timely, considering the growing recognition of mental health and environmental influences on overall well-being, which have often been overlooked in traditional health systems.

[1] Rose Alaslani et al. (2024). *"You can monitor your hydration level using your smartphone camera."*
[2] Singh et al. (2022)—"Fast "*and Robust Video-Based Exercise Classification via Body Pose Tracking and Scalable Multivariate Time Series Classifiers."*
[3] *Souza et al. (2021)*—*"Smartphone "Based Video Analysis for Guiding Shoulder Therapeutic Exercises: Concurrent Validity for Movement Quality Control."*
[4] *Haque et al. (2018)*—*"Measuring "Depression Symptom Severity from Spoken Language and 3D Facial Expressions"*

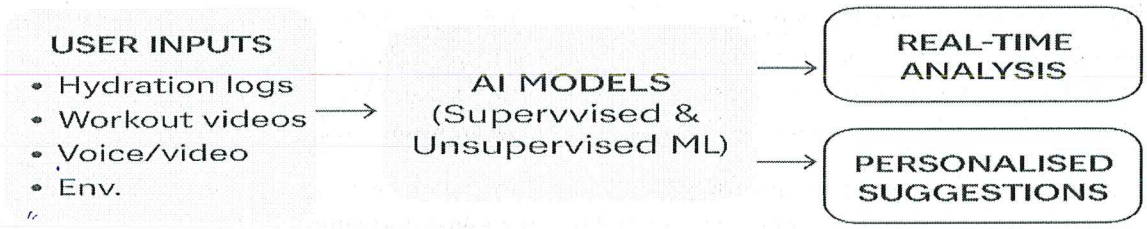
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The suggested solution is a comprehensive health analyser powered by AI that will give real-time evaluations and personalized suggestions for physical, mental, and environmental wellness. This solution uses user-generated content, such hydration logs, workout videos, and voice or facial recordings, as well as real-time environmental data, to give smart insights. This is different from typical systems that rely largely on wearable devices.

The system uses both supervised and unsupervised machine learning algorithms to look at user inputs and change its behaviour based on each person's health profile. For physical health, water needs are predicted in real time based on how active the individual is, the weather, and past data. You may optimize your workouts by using computer vision techniques like object detection and pose estimation to look at exercise movies and provide you immediate feedback and personalized training suggestions.

The technology checks people's emotional states by looking at their facial expressions and tone of voice. It then gives them personalized advice on how to improve their mental health. Environmental health integration also combines outside information (such air quality, weather, and UV index) with how people act to send them health alerts, including when to stay inside.

This AI-powered system makes sure that health optimization is done in a way that is not invasive, flexible, and all-encompassing. It gives each user a smart and personalized experience that improves their overall health.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The AI-driven complete health analyser involves machine learning, data science, health and wellness, computer vision, and environmental science knowledge. Each domain helps the solution provide accurate, adaptive, and effective personalised health advice.

Data science and machine learning expertise:

Supervised Learning: Predicting hydration needs and analysing workout performance using labelled datasets requires expertise in supervised learning methods like Random Forest and Support Vector Machines.

Unsupervised Learning: The system needs clustering and anomaly detection techniques to find trends in mental health and fitness data and make personalised recommendations.

Health & Wellness Expert:

Hydration and Physical Health: Nutrition, exercise science, and physiology experts must establish hydration, workout performance, and physical health metrics. Weather and humidity affect hydration and exercise performance, therefore understanding them is vital.

Mental Health and Stress Management: Clinical psychologists or mental health professionals with experience in emotional regulation and stress management must ensure that mental health data analysis (e.g., facial recognition and voice tone) is accurate and provides meaningful emotional insights.

Image and sound analysis:

Video and Image Processing: Analysing user-uploaded exercise videos for posture and movement feedback requires computer vision skills in pose estimation and object detection.

Vocal tone and pitch can indicate emotional states, however audio signal processing and spoken emotion identification are needed.

Environmental Science and Data Integration:

Environmental health and data analysis skills are needed to integrate air quality and weather data into health optimisation algorithms.

Required Data:

Access to labelled hydration, workout, mental health, and environmental datasets for machine learning model training.

Weather stations or APIs provide real-time data.

User-uploaded video and audio for emotional state detection model training.

8. Objectives and Novelty

Main Objective:The major goal of this study is to create a complete health analyser powered by AI that provides real-time, personalised health optimisation across physical, mental, and environmental health domains. The system uses machine learning algorithms to help with hydration management, workout optimisation, mental health evaluation, and environmental health integration, so you don't have to rely on wearable gadgets as much.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Meyrushan N IT22564818	Hydration Management	-Collect and preprocess user activity logs, environmental data (e.g., temperature), and hydration history. -Use smart bottles to track water levels by capturing timestamped photos of water intake. - Identify hydration patterns such as time gaps, overconsumption, and underhydration by analysing combined data. -Train a supervised machine learning model to predict daily individual water requirements. -Generate weekly or monthly personalised hydration suggestions based on physical activity and environmental conditions. -Alert users about unhealthy hydration trends and predict potential risks such as UTIs or kidney-related issues. -Provide tailored feedback and smart insights to promote optimal hydration and long-term health.	Non-invasive hydration management predicts needs without relying on wearables.

Laxshika S IT22596802	Workout Optimization	- Develop AI algorithms for pose estimation and movement analysis. - Implement Supervised learning to refine workout feedback.	AI-powered video analysis for real-time workout feedback, improving based on user performance.
Kavilakshan V IT22116338	Mental Health Assessment	- Analyze user-uploaded videos for facial recognition and voice tone. - Train models to detect emotional states and provide personalized emotional regulation.	Non-invasive emotional detection using video and audio inputs, personalized over time based on user behavior.
Danushan EJ IT22607546	Environmental Health Integration	- Integrate real-time environmental data (air quality, weather). - Develop models to correlate environmental factors with health outcomes. - Apply Human-Centered Computing (HCC) principles to ensure the data and insights are accessible, meaningful, and usable by individuals.	Proactive health advice based on real-time environmental conditions, personalized to user activities and environment.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Meyrushan N IT22564818	**Hydration Management** : This module uses supervised machine learning to make accurate predictions about how much water a person needs to drink each day depending on their level of activity, the weather (like temperature), and their past drinking habits. It does away with the need for wearable gadgets by using data from user activity records, weather data from the environment, and smart bottles that take timestamped pictures to keep track of how much water a person drinks. The system can tell when someone is drinking too much or too little water by looking at their particular hydration patterns, such as the duration between drinks. To create prediction models that give real-time, non-invasive hydration advice, advanced data science methods like data collecting pipelines, feature engineering, and model training are used. Weekly or monthly trend analysis makes the system's input even better by giving personalized suggestions, like drinking more on hot days or cutting back on overhydration. The device not only gives daily advice, but it also sends out alerts when it sees harmful hydration patterns and can predict health problems like urinary tract infections (UTIs) or kidney problems. This helps people stay hydrated and is good for their health in the long run. This solution is very relevant to IT specialization because it combines important ideas from AI and machine learning, such as predictive analytics, intelligent decision-making, and real-world software deployment, all in the context of a health app that focuses on the user.
Laxshika S IT22596802	**Workout Optimization** : This module employs pose estimation and supervised learning to provide real-time training tips based on video input. It uses AI and computer vision to detect incorrect postures and deliver corrective feedback by comparing user movements to labelled examples of proper form. As the user progresses, the system continues to analyze posture and adjust suggestions based on learned patterns. This approach supports continuous improvement while ensuring accuracy. It aligns with the IT specialization by integrating advanced machine learning models, computer vision techniques, and backend development to create an intelligent and responsive fitness coaching system.

Kavilakshan V IT22116338	**Mental Health Assessment**: This part looks into users' voices, speech tones, and videos they upload to figure out how they feel, like sad, depressed, or happy. To do this, deep learning models are utilized. The technology analyses users' moods and gives them personalized advice to help them control their emotions. This fits with IT fields including Affective Computing, Human-Computer Interaction, and Natural Language Processing. This shows how AI is being used in many areas to figure out and control people's feelings by looking at visual and audio inputs.
Danushan EJ IT22607546	**Environmental Health Integration**: This part uses IoT data integration and machine learning modelling to figure out how factors like air quality and weather affect a person's health. It sends personalized, real-time warnings and health suggestions based on environmental conditions in the user's area. The component fits with IT specialization as it includes data engineering, real-time system architecture, and predictive modelling. It demonstrates how AI can be used to improve public and personal health outcomes through context-aware systems. Additionally, by applying Human-Centered Computing (HCC) principles, the solution ensures that the health insights are understandable, accessible, and actionable for users of varying backgrounds. It emphasizes proactive system design that prioritizes usability, user engagement, and ethical interaction with technology.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr	Kapila	Dissanayaka	Dissanayaka KD 27/06/2025.
Co-Supervisor	Mrs.	Fathima	Fanoon	
External Supervisor				
'Summary of external supervisor's (if any) experience and expertise				

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Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

a) Does the proposed topic exhibit novelty?

Yes	✓	No	

b) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

c) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

d) Supervisor's Evaluation and Recommendation for the Research topic:

Good project to undertake.

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.